Fullstack Product Engineer Case Study

Evidence-Based Fit/Gap Review

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A missing rating is not a skill gap. This change makes the role benchmark, interview evidence, and AI or human rating source visible before a reviewer acts.

Submission Artifact	Access
Open Upstream Pull Request	PR #192: Evidence-Based Fit/Gap Review
Code and Recorded Evidence	Application and Test Snapshot: fd15a9d
Video Demonstration	3:30 Captioned Walkthrough
Successful CI Execution	Run 40207781087: fd15a9d
Seeded Fault and Recovery	Fault: 1ca12dc · Revert: 1426ef5

Before and After

The original table showed a blank Required column and a pencil beside an overridden rating. The revised review shows required L4, effective human L3, original AI L2, and a one-level gap for the same synthetic API requirement. A separate requirement with no usable rating remains **Not assessed**.

Recorded Verification

61 backend examples, **11 frontend tests**, and **4 browser regression tests** pass. Verification also covers a frontend production build, a scoped mobile accessibility scan, and migration down/up with existing rows. A deliberate arithmetic fault caused **28 of 35 focused examples** to fail before its revert.

Evidence Boundaries

Candidate names, quotes, notes, and accounts shown are synthetic fixtures. Simulated UI failures are labeled. The live microphone/Gemini interview was not verified. AI assistance was used for analysis, implementation, tests, and report preparation; the specific checks and remaining limits are documented below.

Application/Test Evidence Snapshot

Baseline: b836d02f45c6300b238b98ad50e66ead706723b6
Verified application and tests: fd15a9d8a99090ec46c1c31ad28e819bf5a64286
This report revision changes documentation and figure presentation; it does not claim new application test coverage.

A Blank Benchmark at the API/UI Seam

The supplied repository was cloned from main, both service READMEs and the product wiki were read, and a feature branch was created. Rails, PostgreSQL, Redis/Sidekiq, React, TypeScript, and Vite were retained. Docker in Ubuntu WSL supplied Ruby 3.3.2 and Redis without altering the existing Windows PostgreSQL installation.

Vite runs on port 5174 because 5173 was occupied; the API uses 3001. A development-only, rerunnable seed provides synthetic assessment data. The selected comparison path requires no provider key.

Baseline Reproduction

Skill	Required	Candidate	Result
Database Design & SQL		L3	✅ Match
RESTful API Design		L3	⚠️ Gap -1
Testing & Quality Assurance		L4	⭐ Exceeds +2
Communication		—	— Not assessed

Original frontend with modified backend, synthetic data. The Required column is blank. This is a reproduction of the presentation defect, not an untouched whole-system baseline. Baseline Frontend Capture

The baseline engine emits **expected_level**; the original table reads **required_level**. The first frontend regression run failed all three checks for benchmark/provenance rendering, missing-rating clarity, and empty requirements. Original Failing Test Log

Exploration Scope and Timing

Browser exploration covered login, assessment list, a completed synthetic session, portfolio evidence, vacancy selection, comparison, refresh, transcript, and downloads. API payloads and database relationships were inspected. Backend lifecycle work had already started when the old frontend was reproduced; the full live interview was not exercised before implementation. That departure from the requested discovery sequence remains a limitation.

Evidence for Reviewers and Candidates

The PRDs describe an AI-led interview, transcript-backed skill evidence, and a vacancy comparison. The selected outcome is a defensible follow-up decision: identify what meets the benchmark, what falls below it, and what was never established. The roles below are hypotheses from the brief and PRDs, not user-interview findings.

Person	Review Need	Design Consequence
Assessor	Inspect the evidence and correct an AI rating.	Show quotes, notes, original AI level, and effective human level.
Recruiter	Compare observed skills with the role benchmark.	Place required and observed levels together; separate missing evidence from gaps.
Hiring Manager	Decide what needs another conversation.	Filter gaps, unassessed requirements, and low-confidence AI ratings.
Candidate	Avoid an unsupported or uncorrectable negative assessment.	Do not invent L1, erase a human review, or expose another tenant's data.

An Indonesian Hiring Context

Kemnaker's Pasker portal links to Karirhub and already displays public vacancies. It also publishes vacancy datasets by education, occupation, and experience requirements. These are observable discovery and eligibility services. This assessment tool needs to add evidence of demonstrated skill beyond a listing or eligibility field.

Design inference: prioritize required-versus-observed levels and correctable evidence over another generated hiring verdict. This does not establish that every competing assessment product lacks those features. A useful next check is whether recruiters can explain a follow-up decision from the evidence without treating an unassessed skill as a failure.

Candidate Privacy and UU PDP

UU 27/2022 covers correction (Article 6), access/copies (Article 7), objections to solely automated decisions with legal or significant effects (Article 10), and security/confidentiality (Articles 35-36). The design supports inspection, correction, and tenant boundaries. Retention, notices, legal bases, and provider agreements require separate work; no compliance certification is claimed.

Sources and Research Timing

PRD 01: Evidence and Confidence · PRD 02: Interview and Review
Kemnaker Pasker: Services and Vacancy Datasets · UU 27/2022: Official Text
PRD/privacy context was documented on 25 October. The more specific public-service comparison above was added during the 26 October report review.

Assessment Integrity Takes Priority

P0: broad exposure/unusable system. P1: incorrect assessment or unauthorized disclosure. P2: recoverable reliability/workflow issue. P3: minor clarity/polish. These are qualitative priorities, not measured incident probabilities.

ID / Severity	Finding and User Impact	Type / Implemented Response
F1 · P1	Required levels disappear; reviewers cannot inspect the benchmark behind a gap.	Defect at the API/UI seam. One expected_level contract.
F2 · P1	Rating provenance disappears; reviewers cannot distinguish AI output from human judgment.	Serialization defect. Include original/effective levels, source, and notes.
F3 · P1	Child lookups omit tenant ownership; a signed-in assessor can attempt foreign portfolio/skill IDs.	Controller defect. Owned session joins and vacancy checks.
F4 · P2	Repeated generation and unrecorded failures can strand a reviewer or discard an old report.	Defect / missing lifecycle specification. Tokens, status, timeout detection, and retry.
F5 · P1	Model ratings are coerced into 1..5 and skills deleted before validation; ratings or reviews can be wrong or lost.	Generator defect. Strict validation, atomic writes, retained human reviews.
F6 · P1	Culture fit and hiring recommendations are requested without grounded evidence.	Missing evidence policy. Deterministic summary; no culture/hiring inference.
F7 · P3	Incomplete summaries and missing edge states make follow-up harder.	Defect / missing state design. Completeness counts and recovery states.

Priority Revision

The initial plan assigned F6 P2. The final report treats it as P1 because an unsupported hiring recommendation could affect candidate evaluation, rather than merely interrupt a recoverable workflow. This is a revised risk judgment, not a measured incident or a retroactive change to the initial plan.

Early Constraint Signal

The original interview depends on live Gemini credentials and audio. Shared tenant/auth assumptions and the worker contract require review. The chosen comparison can operate without a provider, allowing one end-to-end review workflow within the four-day constraint. The initial plan records these dependencies before feature implementation. Initial Discovery and Acceptance Plan

A Deterministic Review within the Existing Stack

The original plan compared a frontend field repair, a persisted deterministic comparison, and replacement of the interview engine. The selected change addresses the benchmark defect together with provenance, authorization, refresh correctness, and failure recovery.

Original Option	Value and Work Required	Cost, Failure, and Decision
A. Field Repair	Correct the frontend field with minimal API change.	Quick benchmark visibility; leaves stale snapshots, lost reviews, and tenant lookup defects. Rejected as too narrow.
B. Persisted Review	Add lifecycle metadata, snapshot inputs, worker guards, and review states.	API/web maintainers own the digest and retry policy. Additive migration is reversible. Selected; reviewer-driven retry is still required.
C. Interview Replacement	Rework model behavior, live audio, and the generation pipeline.	Expands provider and real-time failure surfaces. Cannot establish reliable evidence within this scope. Deferred.

Closer Alternative: Queue Recovery

The implementation chose commit-before-enqueue with timeout detection and explicit retry instead of an outbox. The comparison below makes that operational trade-off explicit during this report revision; it is not presented as an earlier experiment.

Approach	Operation and Maintenance	What It Gives Up
Selected: Token + Retry	Keep the existing Rails/Sidekiq stack. A two-minute threshold exposes stuck work; a reviewer requests a new attempt. Maintainers own the timeout and recovery copy.	A crash after commit can leave no queued job. No automatic re-enqueue or fixed recovery-time guarantee.
Alternative: Outbox	Persist a dispatch event with the report transaction. Add an idempotent dispatcher, retry/backoff policy, and monitoring; the backend team owns the new process.	More schema and operational work. Still needs idempotent workers and queue recovery. Rollback must account for undelivered events.

Why the Selected Trade-Off Fits

The review remains usable with a preserved snapshot and visible retry while requiring fewer new moving parts. If abandoned pending reports or unattended recovery become material, an outbox offers a specific benefit worth the added maintenance. No workload threshold or saved engineering hours has been measured.

Defined Behavior for Missing and Changed Evidence

Acceptance criteria were recorded before the feature changes. The implementation checks below organize them by observable behavior; the original plan remains available without rewriting its history. Pre-Implementation Plan

1. Benchmark. Compare every vacancy requirement in stable order. Show required/effective levels and a signed delta.

2. Identity. Use normalized canonical IDs. Restrict label fallback to custom skills; ambiguous duplicates remain unassessed.

3. Missing Ratings. Use not_assessed and no numeric delta for a missing/invalid rating. Never fabricate L1 by coercion.

4. Human Review. Retain original AI level, effective level, source, notes, and review time. Changed inputs invalidate cached comparisons.

5. Authorization. Require portfolio/session and vacancy tenant ownership for review, export, override, and worker execution. Reject missing auth, wrong roles, and foreign IDs.

6. Job Lifecycle. Deduplicate active requests and deliveries with status/token checks. Expose an old pending/generating attempt as timed out after two minutes; recovery requires explicit retry.

7. Saved Content. Preserve the last successful snapshot during refresh failure or partial writes. Block comparison exports while stale or processing.

8. Decision Limits. Keep comparison deterministic and provider-independent. Do not infer culture fit or issue an automatic hiring recommendation.

9. Data and UI Safety. Use an additive reversible migration. Handle loading, empty, error, partial, long-text, narrow-screen, and keyboard states.

10. Proof. Exercise backend/frontend/browser regressions, seed and revert a fault, and document a checked/corrected AI contribution.

Scoring Assumptions

L1-L5 are ordered categories, not a calibrated interval score. A signed level difference supports follow-up; it is not an overall fit percentage. A complete portfolio may still leave some vacancy requirements unassessed. A human override remains the effective rating until explicitly revised.

Versioned Inputs and Atomic Report Writes

FitGap::Snapshot reads ordered requirements, ratings, quotes, summaries, and override provenance. A SHA-256 digest identifies the inputs behind a successful comparison. Canonical identity avoids matching unrelated skills merely because their labels are the same.

Lifecycle Field	Purpose
generation_status	pending, generating, complete, failed, or stale. Existing rows without a digest require refresh.
generation_token / requested_at	Identify an attempt and its timeout threshold. An obsolete delivery cannot replace a newer request.
source_digest	Detect changed requirements or review inputs on reads and exports.
error_code	Safe queue_unavailable, generation_failed, or derived generation_timeout. No provider payload is stored here.

Request and Worker Sequence

1. Resolve the tenant and owned portfolio/vacancy.
2. Lock the portfolio; reuse a current/active report or save a pending token.
3. Commit before enqueueing the Sidekiq job.
4. Worker resolves ownership, takes the same lock, and checks the token.
5. Compute comparisons and atomically save the successful snapshot.
6. On worker exception, roll back partial data and record a safe failure.

Timeout Detection and Recovery

After two minutes, an outstanding attempt is exposed as failed when its status is read. This does not automatically re-enqueue it or stop a running process. A reviewer must explicitly retry; completion depends on that action and queue availability. An enqueue exception retains the previous successful comparison. Exports require a current digest.

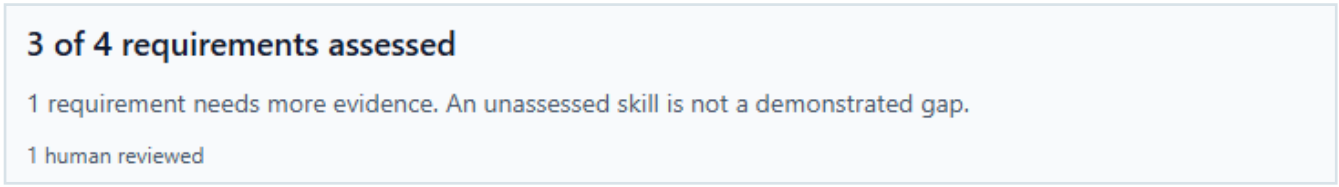
Portfolio Validation and Data Safety

Missing levels stay absent; malformed, non-integer, and out-of-range levels fail validation. Invalid response fragments roll back the write. Matching skills retain human overrides, and human-reviewed skills are kept if a provider omits them. Migration down/up preserved existing comparisons and owning rows; restored legacy reports were stale until refreshed. Migration Check

Required Levels and Human Rating Sources

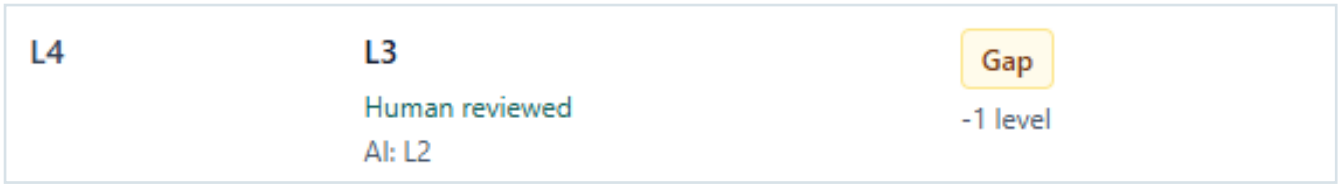
The review extends the existing light Rakamin teal system: completeness first, benchmark and observed rating second, supporting evidence on demand. Text accompanies status colors. The captures below use synthetic data and retain the application's original text casing.

Assessed Requirements Stay Visible



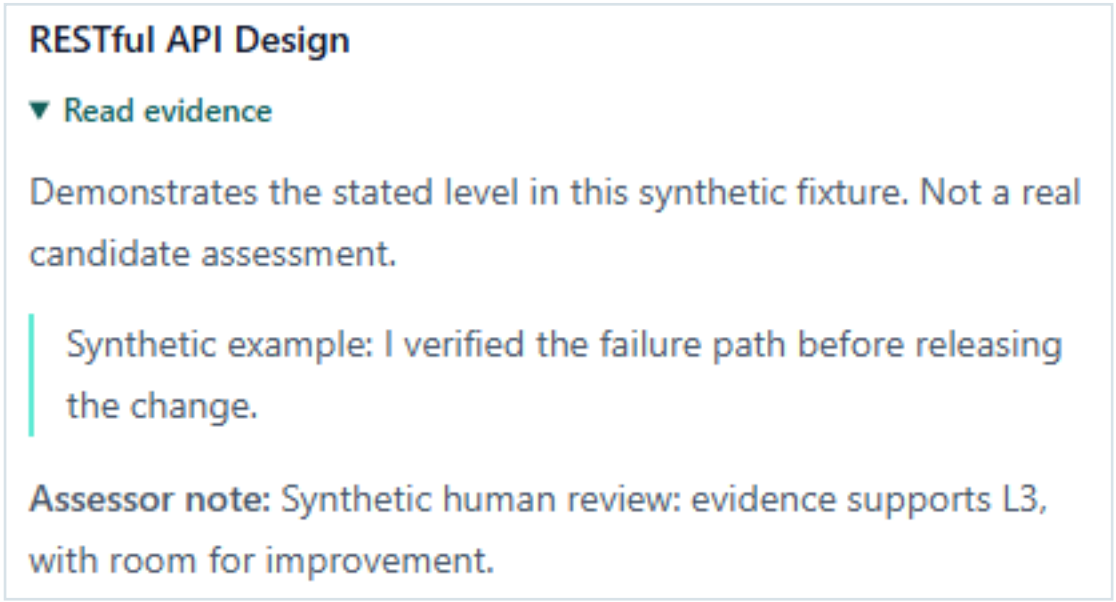
Three of four requirements assessed. One requirement needs more evidence; the count does not call it a demonstrated gap.

Human Review Retains the Original AI Level



RESTful API Design, left to right: required L4, effective human L3, original AI L2, and a one-level gap. These are synthetic acceptance-fixture values.

Quotes and Notes Explain the Revision



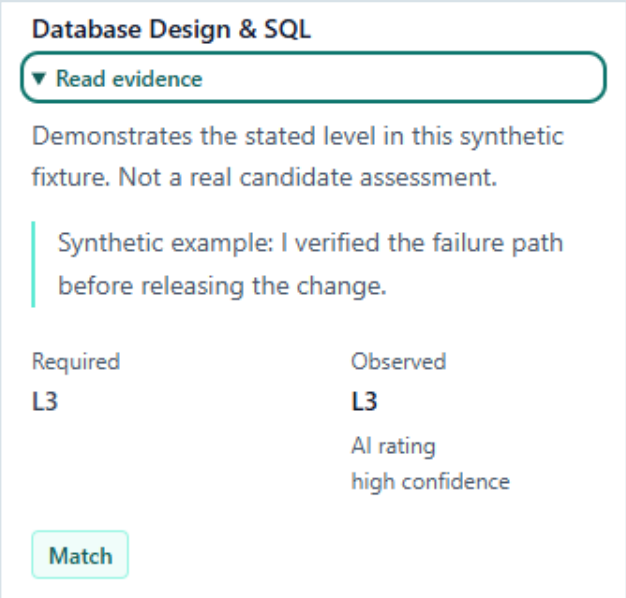
Synthetic evidence and assessor note. The note explains the L3 revision while the comparison keeps the original AI source available.

Desktop Review: Full Capture

Unassessed Skills Remain Distinct from Gaps

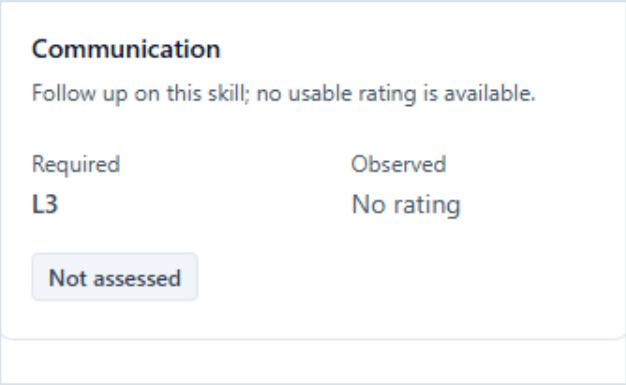
On narrow screens, requirement rows become stacked cells. The two details below come from the same 375-pixel capture and are enlarged for reading inside this PDF.

Keyboard Evidence Disclosure



Synthetic mobile fixture. Enter opens this focused disclosure. This verifies activation of that control, not full keyboard traversal.
Mobile Review: Full Capture

Missing Ratings Stay Unassessed



Communication requires L3 but has no observed rating. It remains Not assessed, with no numeric delta.

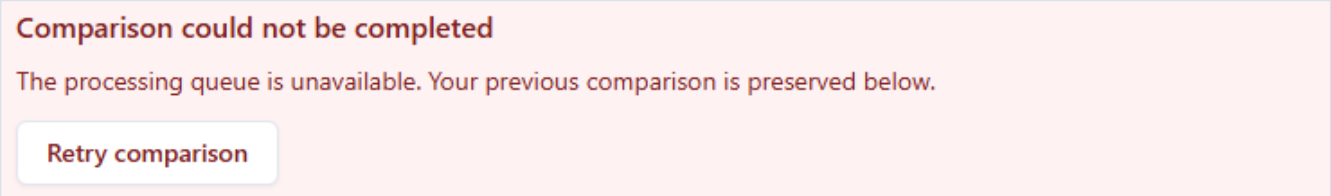
Scoped Accessibility and Long Text

The mobile browser check programmatically focuses an evidence control and presses Enter, then runs an axe WCAG A/AA scan. The scoped scan has no reported violations after the active-navigation contrast fix. A separate 320-pixel test checks an unbroken long label for document overflow. Neither check certifies the whole product or PDF accessibility.

Previous Evidence Survives a Failed Refresh

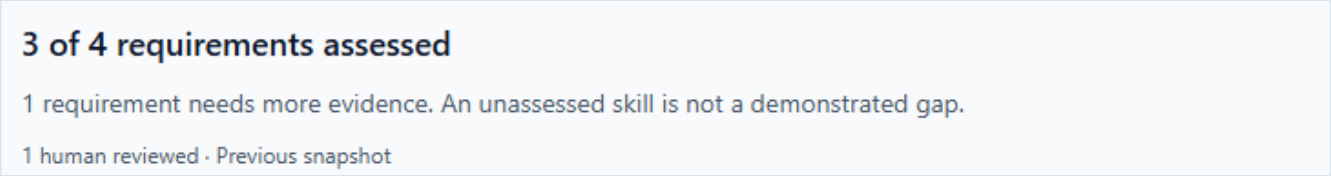
These details show an explicitly simulated queue-failure API response for UI testing. They demonstrate the visible failure state; they are not footage of a real queue outage.

A Visible Error and an Explicit Retry



Simulated queue failure. The message names the unavailable queue, confirms preservation, and offers Retry comparison.

The Saved Snapshot Keeps Its Status



Previous snapshot. The assessed count and human-review context remain visible, with the snapshot identified as previous.

Exports Stay Disabled Until the Report Is Current



Refresh remains available; PDF and JSON are disabled. This prevents downloading a comparison presented as current after refresh failure.

Simulated Refresh Failure: Full Capture

Other Tested States

Frontend tests cover initial-load failure/retry, a pending portfolio, stale exports, refresh failure, and download failure with Check status recovery. Browser response simulations also cover long labels and an empty vacancy.
Frontend Lifecycle Checks · Browser State Checks

Regression Evidence with a Defined Scope

Check	Recorded Result and Evidence
Backend	61 examples, 0 failures: all 25 level pairs, tenant denial, malformed inputs, identity, provenance, duplicate jobs, safe failures, and rollback. Backend Log
Frontend	11 tests pass; TypeScript/Vite production build succeeds. Frontend Log · Build Log
Browser	4 regression checks: real API evidence/filter/refresh/download/transcript flow; narrow-screen Enter/axe checks; labeled failed-refresh, long-label, and empty simulations. Browser Log
Migration	Down/up preserves comparisons and owning rows; restored legacy report requires refresh. Migration Script
Dependencies	Recorded production npm audit: 0 advisories. Inherited development-tool advisories are not represented as cleared. Production Audit JSON

Interaction Claims Match the Tests

Browser checks operate disclosure, filtering, refresh/retry, PDF/JSON download, and transcript navigation. Frontend checks exercise initial-load retry and Check status recovery. The recorded walkthrough enters from the portfolio. Return-to-portfolio links and a complete Tab/Shift+Tab route are not claimed as tested by those records. Exact Browser Actions

Seeded Fault and Revert

On a scratch branch, subtraction was replaced by addition. The focused suite produced 28 failures in 35 examples. The fault was committed and reverted; assertions were not weakened. Fault 1ca12dc · Revert 1426ef5 · Failure Log

AI Verification and Correction

An AI-generated browser selector initially missed an evidence control whose accessible label includes skill-specific text. It was corrected to target the summary element while retaining the behavior assertion. The reviewable test focuses that control, presses Enter, and asserts that details is open; it does not prove full keyboard traversal. During this report review, the broader AI-authored all-controls claim was removed after checking those exact actions. Verified Disclosure Assertion

A dependency review also replaced the initially selected Vitest 3.2.x with 5.0.5. The original audit output was not retained, so this revision does not assert exact historical package/advisory severity. The committed lockfile and successful CI establish the final versions and checks. Dependency Lockfile · Exact Successful CI Run

Submission Artifacts and Remaining Work

Submission Artifacts

Open Upstream PR #192
3:30 Captioned Video Demonstration
Revised Report Source and Documentation
Verified Application and Test Snapshot
Successful CI Run for That Snapshot

The video contains an actual local API/worker browser recording using synthetic fixtures, slowed for readability. Queue-failure footage is labeled as a response simulation. It has explanatory on-screen text and no spoken narration. This PDF is the single portal document; external links provide code, proof, and video.

Material Limits

• The live microphone/Gemini interview was not exercised. Provider validation/failure paths use controlled clients; the comparison itself makes no model call.

• The two-minute threshold detects old attempts; it does not dispatch lost jobs. Explicit reviewer retry and queue availability determine recovery. An outbox remains an alternative.

• Protection is scoped to report, export, and override paths. Whole-product auth/WebSocket auditing, rights-request workflows, retention, and provider contracts remain separate work.

• Previously coerced AI levels cannot be reconstructed without original responses. Current override storage keeps the current review, not an immutable revision ledger.

• No production-load, time-saving, fairness, or hiring-accuracy improvement has been measured. Synthetic acceptance tests establish behavior, not business effectiveness.

Release and Rollback

Apply the additive migration before the new code. Drain legacy two-argument fit/gap jobs before switching to the three-argument worker. Rollback preserves comparison/portfolio data but removes lifecycle metadata. The local-only Compose runtime contains documented synthetic credentials; production configuration is outside this change.

Runbook and Defense Preparation

Setup, Checks, and Rollback Runbook
Separate Technical Defense Preparation